

478 Project

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Abstract—This project investigates whether incorporating weather conditions improves the prediction of cab ride fares. We made two linear regression models: a baseline model using only ride features and an extended model that also includes weather variables such as temperature, humidity, and wind. The dataset was cleaned and split into 80% training and 20% testing data. Both models achieved nearly identical performance, with an R^2 score of 0.9287. Error analysis showed similar stats, with a mean error of about -0.0019 and variance near 6.19 for both models. These results indicate that ridespecific variables already capture most of the variation in ride prices, and weather data does not significantly improve predictive performance. Therefore, while linear regression is effective for this task, additional environmental features do not provide meaningful benefits in this dataset.

I. INTRODUCTION

The purpose of this project is to examine whether weather conditions improve the prediction of cab fares. These fares are influenced by several factors, like distance, time, demand, and ride type, but external factors such as temperature, rain, humidity, and wind may also affect pricing patterns. To investigate this, we combined ride and weather data to build two separate multiple linear regression models. The first only used ride variables, like distance and surge multiplier. The second used these alongside weather features, like temperature and pressure. After cleaning and removing rows without required values, the dataset was split into 80% data and 20% testing to evaluate how well each model would predict the price.

The results showed that both models performed very strongly, but weather features did not affect prediction accuracy. The first model produced an R^2 score of .9287, while the second model, including weather, also produced an R^2 score of .9287. The residual error statistics were close to identical as well, with a mean error of about -.0019 and an error variance of about 6.19. Based on these results, we can conclude that linear regression is effective for modeling ride prices for this dataset, but ride specific variables already explain almost all of the variation in price, and the added weather data does not provide meaningful predictive value.

II. PROBLEM

The problem addressed in this project is a supervised regression task in which the goal is to predict the price of a cab ride from a set of observed features. The output of the model is the continuous target variable price, which represents the fare of an individual ride. The inputs consist of ride-related features and, in one version of the model, weather-related features. The ride inputs include numerical variables such as distance, surge_multiplier, hour, and day_of_week, along with categorical variables such as cab_type, destination, and product_id, which were encoded as dummy variables for use in linear regression. The extended model also includes weather inputs matched to each ride, specifically temp, clouds, pressure, rain, humidity, and wind. Formally, the task is to learn a function that maps these input features to a predicted ride price as accurately as possible on unseen test data.

Going into this project, we wanted to find out if including weather conditions as part of our input feature set would improve the prediction of cab prices. The output of our models is the continuous target variable price, which is just the fare of any individual ride. Our inputs consist of ride features in both models, while the second contains weather features as well, as a point of comparison. The ride inputs include numerical variables such as distance, hour, and day_of_week, alongside categorical variables such as cab_type, destination, and product_id, which were encoded as dummy variables for use in linear regression. The weather features included in the extended model are features matched to each ride such as temp, clouds, rain, and wind. We set the task for our models to predict ride price based on these inputs and compare to unseen test data set aside at the start.

Our analysis was done by creating two multiple linear regression models to estimate the ride price and compare the effect of adding environmental information. The baseline model only uses ride specific features, while the second contains environmental measurements. We chose linear regression because it is a standard, interpretable method for continuous prediction and gives a direct way to measure how much each added group of variables contributes to the overall fit. Making these two models so similar and closely related allows the

comparison to be controlled. Both models share the same standard structure and only differ in whether certain features are included allowing us to closely examine the effect of those features, as any difference in predictive performance can be attributed to that specific additional info rather than to a change in the model family. This setup is well suited to testing our main research question, which is whether weather data makes predicting cab fares easier.

III. PERFORMANCE INDICATORS AND JUSTIFICATION

To evaluate the effectiveness of our models, we used the coefficient of determination (R^2) as the primary performance metric, along with error statistics including mean error and variance. The R^2 score measures how well the model explains the variance in the target variable, making it a standard choice for regression problems. Additionally, examining the mean and variance of prediction errors provides some good insight into bias and consistency. Both the baseline ride only model and the other ride and weather model achieved identical R^2 scores of 0.9287 on the test dataset. This indicates that both models explain approximately 92.87% of the variation in ride prices. Furthermore, the residual error statistics were nearly identical between the two models, with a mean error of approximately -0.0019 and a variance of about 6.19. These results show that both of these models produce accurate and unbiased predictions at about the same level.

IV. PROS AND CONS

One major advantage of this approach is its simplicity and readability. Linear regression provides a clear relationship between input features and predicted price, allowing us to understand how variables such as distance and surge multiplier affect fares. Additionally, using two closely related models enabled a controlled comparison, making it easier to study the weather features by itself. The high R^2 score also shows that the chosen features are effective in explaining ride price variation. Another strength is the use of real-world data because this ensures that the data being used is clean and meaningful. However, there are also limitations to this approach. Linear regression assumes a linear relationship between inputs and output, which may not fully show any other different type of relationship that might not be linear. Additionally, the lack of improvement from weather features may come from how the data was matched for the weather data. More advanced models, such as decision trees or neural networks, might better capture these different relationships. Finally, while the dataset is large and informative, it may not include other cities or time periods that could greatly change the results. External factors such as special events or traffic conditions were not included and could also influence pricing. Given that the extended model does not improve performance despite including additional features, the ride only model is preferable. It achieves the same predictive accuracy while being simpler and more readable. This is great because some models have better accuracy when they are simple.